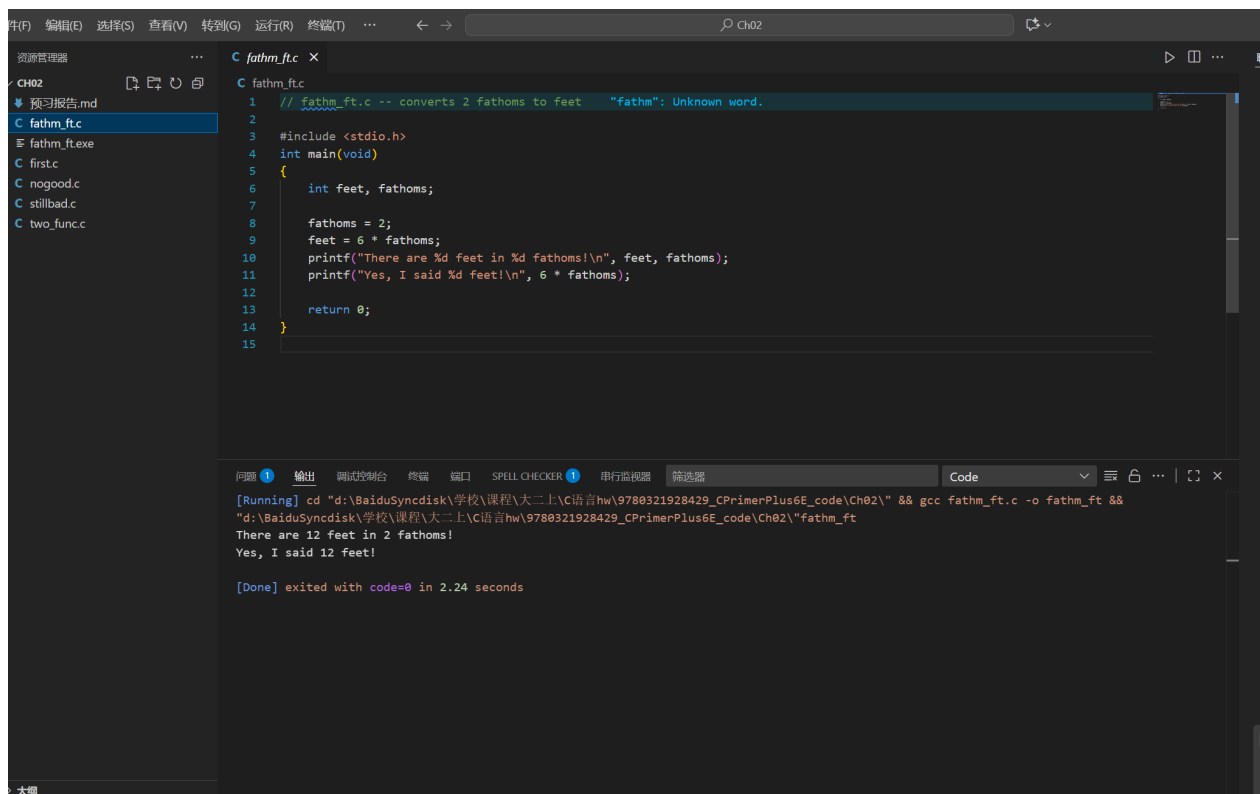


第2章预习报告

1. fathm_ft.c

- **作用:** 英寻到英尺的转换程序
- **功能:** 将2英寻转换为英尺（1英寻=6英尺）
- **演示效果:**



The screenshot shows a code editor with a file explorer on the left and a terminal window at the bottom. The file explorer lists files: CH02, 预习报告.md, C fathm_ft.c, fathm_ft.exe, C first.c, C nogood.c, C stillbad.c, and C two_func.c. The code editor displays the source code for fathm_ft.c:

```
1 // fathm_ft.c -- converts 2 fathoms to feet "fathm": Unknown word.
2
3 #include <stdio.h>
4 int main(void)
5 {
6     int feet, fathoms;
7
8     fathoms = 2;
9     feet = 6 * fathoms;
10    printf("There are %d feet in %d fathoms!\n", feet, fathoms);
11    printf("Yes, I said %d feet!\n", 6 * fathoms);
12
13    return 0;
14 }
15
```

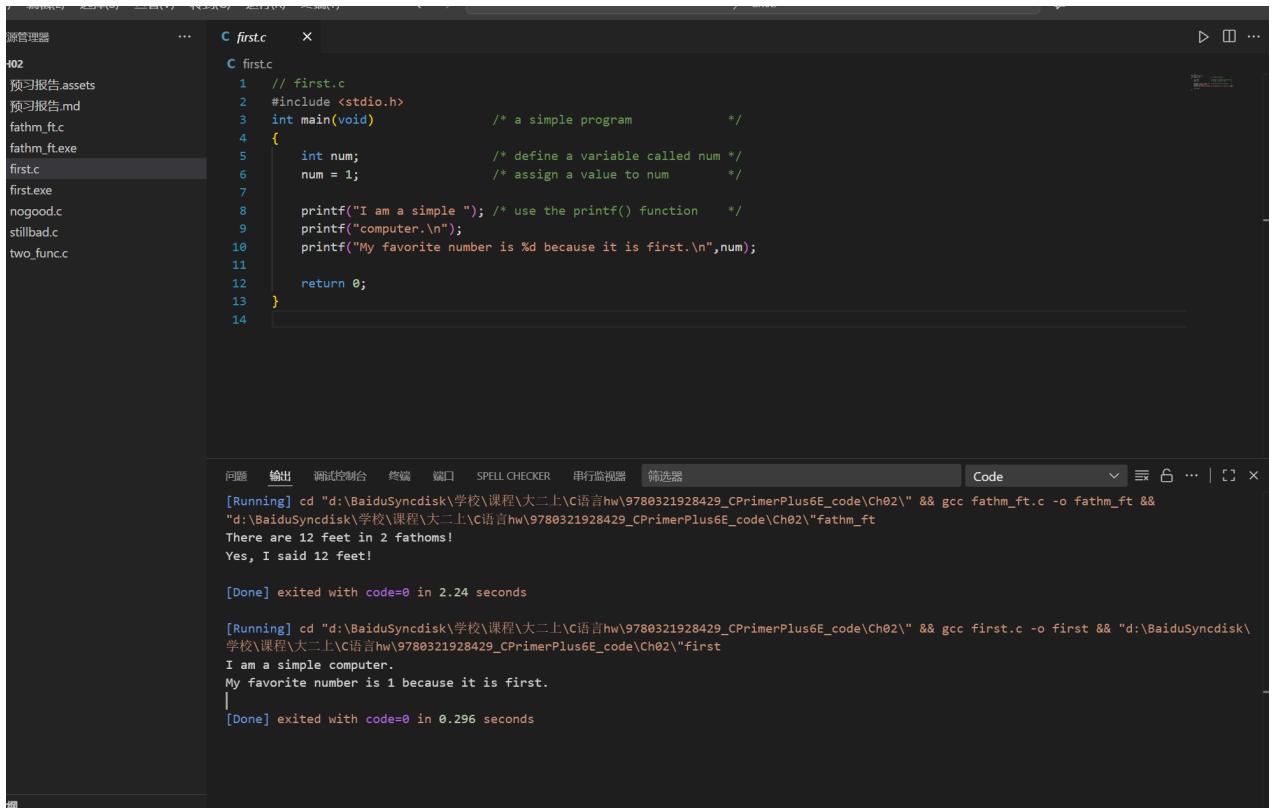
The terminal window shows the execution output:

```
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc fathm_ft.c -o fathm_ft &&
"d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\"fathm_ft
There are 12 feet in 2 fathoms!
Yes, I said 12 feet!

[Done] exited with code=0 in 2.24 seconds
```

2. first.c

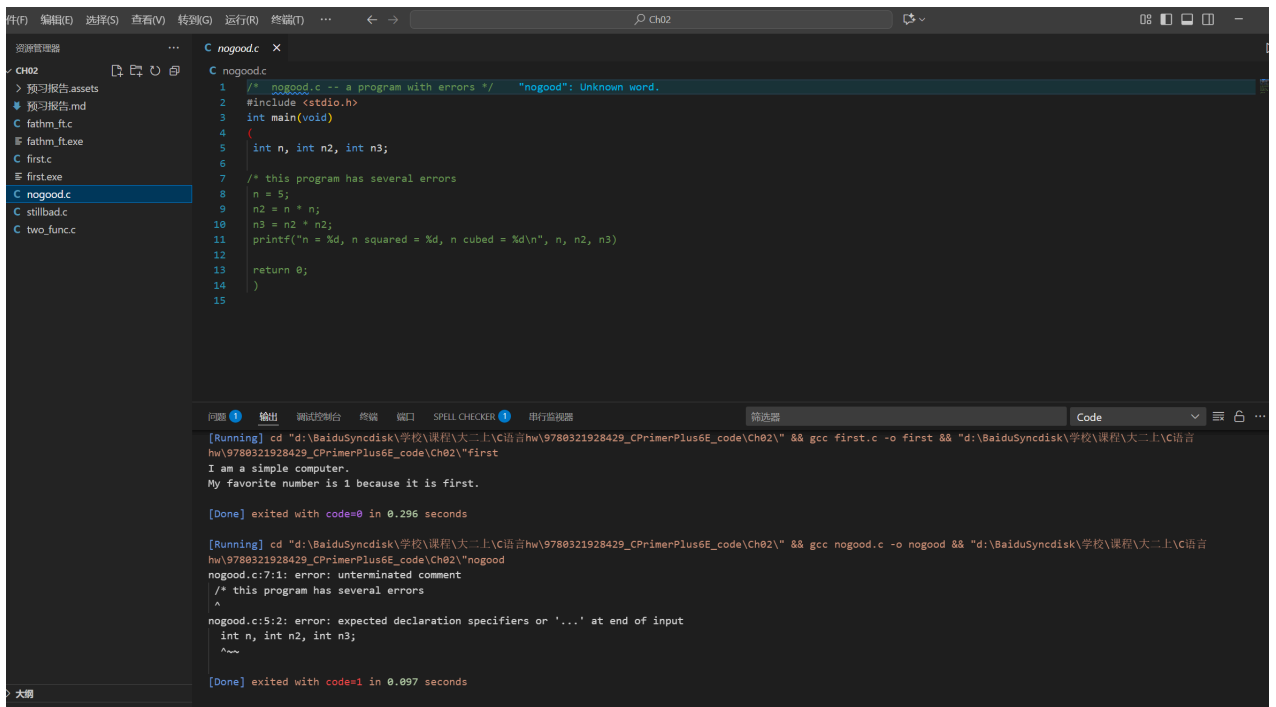
- **作用:** 简单的C语言入门程序
- **功能:** 输出简单的文本信息和变量值
- **演示效果:**



3. nogood.c

- **作用:** 包含语法错误的示例程序
- **功能:** 本意是计算n的平方和立方，但存在多个语法错误
- **主要错误:** 使用圆括号()而不是花括号{}、变量声明语法错误、缺少分号等
- **演示效果:**

刚开始编译错误:



修正后结果:

```
1  /* nogood.c -- a program with errors */      "nogood": Unknown word.
2  #include <stdio.h>
3  int main(void)
4  {
5      int n, n2, n3;
6      // this program has several errors
7      n = 5;
8      n2 = n * n;
9      n3 = n2 * n2;
10     printf("n = %d, n squared = %d, n cubed = %d\n", n, n2, n3);
11     return 0;
12 }
13
```

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc nogood.c -o nogood && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\"nogood
nogood.c:7:1: error: unterminated comment
/* this program has several errors
^
nogood.c:5:2: error: expected declaration specifiers or '...' at end of input
int n, int n2, int n3;
 ^~~~
[Done] exited with code=1 in 0.097 seconds
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc nogood.c -o nogood && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\"nogood
n = 5, n squared = 25, n cubed = 625
[Done] exited with code=0 in 0.331 seconds

4. stillbad.c

- **作用:** 修正了nogood.c语法错误但仍有语义错误的程序
- **功能:** 计算n的平方和立方（但逻辑有误）
- **语义错误:** n3应该是n的立方，但代码中写成了n2的平方
- **演示效果:**

刚开始逻辑错误:

```
1  /* stillbad.c -- a program with its syntax errors fixed */      "stillbad": Unknown word.
2  #include <stdio.h>
3  int main(void)
4  {
5      int n, n2, n3;
6
7      /* this program has a semantic error */
8      n = 5;
9      n2 = n * n;
10     n3 = n2 * n2;
11     printf("n = %d, n squared = %d, n cubed = %d\n", n, n2, n3);
12
13     return 0;
14 }
15
```

nogood.c:5:2: error: expected declaration specifiers or '...' at end of input
int n, int n2, int n3;
 ^~~~
[Done] exited with code=1 in 0.097 seconds
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc nogood.c -o nogood && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\"nogood
n = 5, n squared = 25, n cubed = 625
[Done] exited with code=0 in 0.331 seconds
[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc stillbad.c -o stillbad && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\"stillbad
n = 5, n squared = 25, n cubed = 625
[Done] exited with code=0 in 0.338 seconds

后面修改正常:

```
1 /* stillbad.c -- a program with its syntax errors fixed */    "stillbad": Unknown word.
2 #include <stdio.h>
3 int main(void)
4 {
5     int n, n2, n3;
6
7     /* this program has a semantic error */
8     n = 5;
9     n2 = n * n;
10    n3 = n2 * n;
11    printf("n = %d, n squared = %d, n cubed = %d\n", n, n2, n3);
12
13    return 0;
14 }
15
```

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc nogood.c -o nogood && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\nogood"
n = 5, n squared = 25, n cubed = 125

[Done] exited with code=0 in 0.331 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc stillbad.c -o stillbad && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\stillbad"
n = 5, n squared = 25, n cubed = 125

[Done] exited with code=0 in 0.338 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc stillbad.c -o stillbad && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\stillbad"
n = 5, n squared = 25, n cubed = 125

[Done] exited with code=0 in 0.343 seconds

5. two_func.c

- 作用: 演示多函数使用的程序
- 功能: 主函数调用butler函数，展示函数声明和定义
- 演示效果:

```
1 /* two_func.c -- a program using two functions in one file */
2 #include <stdio.h>
3 void butler(void);    /* ANSI/ISO C function prototyping */
4 int main(void)
5 {
6     printf("I will summon the butler function.\n");
7     butler();
8     printf("Yes. Bring me some tea and writeable DVDs.\n");
9
10    return 0;
11 }
12
13 void butler(void)    /* start of function definition */
14 {
15     printf("You rang, sir?\n");
16 }
17
```

n = 5, n squared = 25, n cubed = 125

[Done] exited with code=0 in 0.338 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc stillbad.c -o stillbad && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\stillbad"
n = 5, n squared = 25, n cubed = 125

[Done] exited with code=0 in 0.343 seconds

[Running] cd "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\" && gcc two_func.c -o two_func && "d:\BaiduSyncdisk\学校\课程\大二上\C语言hw\9780321928429_CPrimerPlus6E_code\Ch02\two_func"
I will summon the butler function.
You rang, sir?
Yes. Bring me some tea and writeable DVDs.

[Done] exited with code=0 in 0.318 seconds